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Supplemental information

**The effect of neural network
architecture on virtual H&E staining: Systematic
assessment of histological feasibility**

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Supplemental Figures

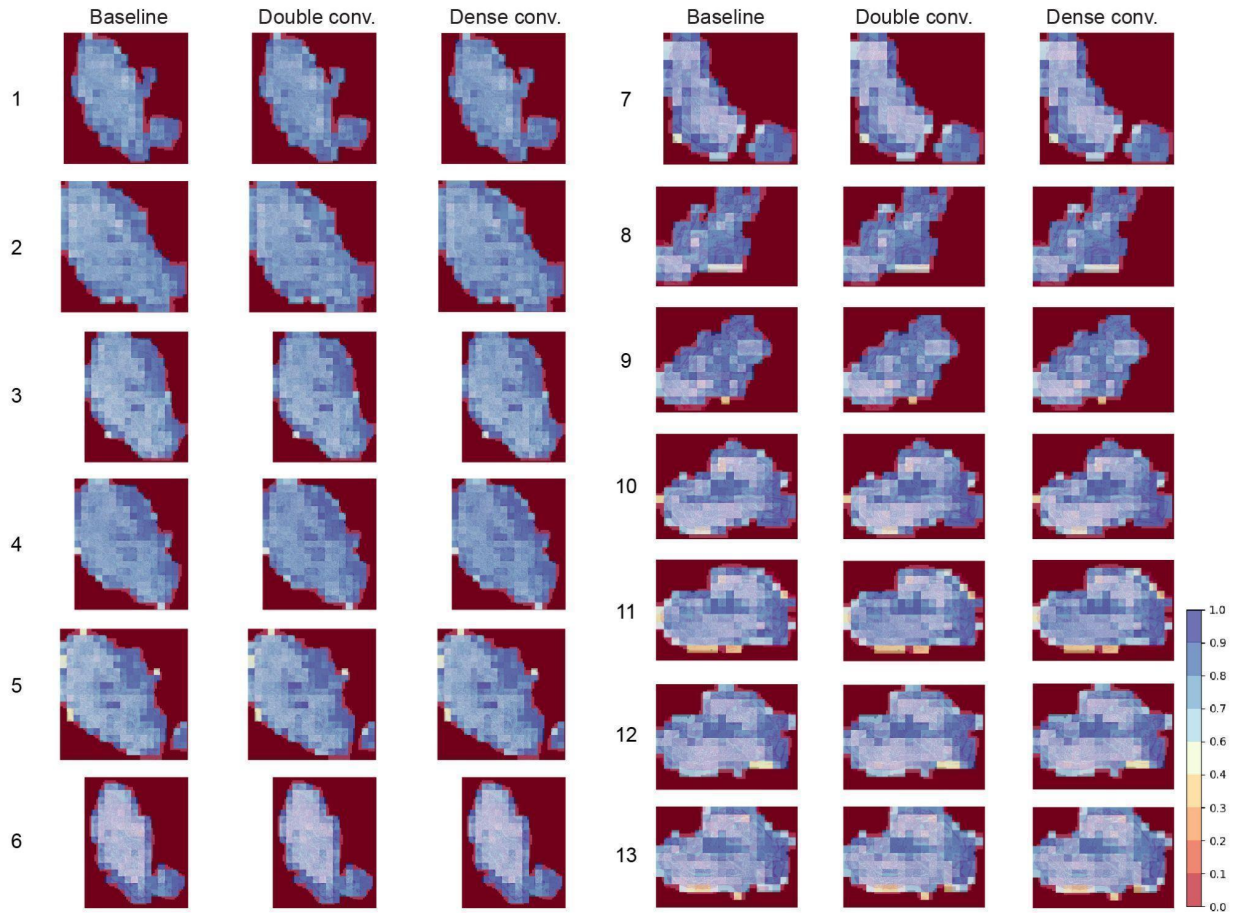


Figure S1. Tile-wise SSIM heatmap overlay. A collage of low-resolution tissue images in the test set overlaid with tile-wise SSIM score. The color bar represents the SSIM intensity between 0 and 1, the bluer tile represents high similarity between virtually stained tile and its corresponding ground truth tile.

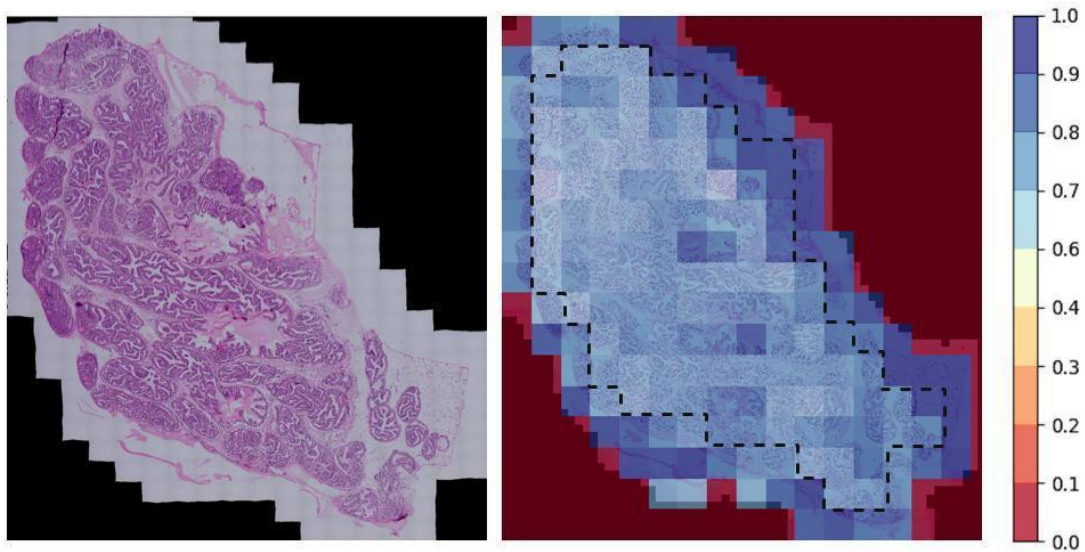


Figure S2. SSIM heatmap-based edge tile exclusion. An example tissue image in the test set overlaid with tile-wise SSIM scores (similar to Figure S1), showing the excluded edge areas outside the dotted boundary.

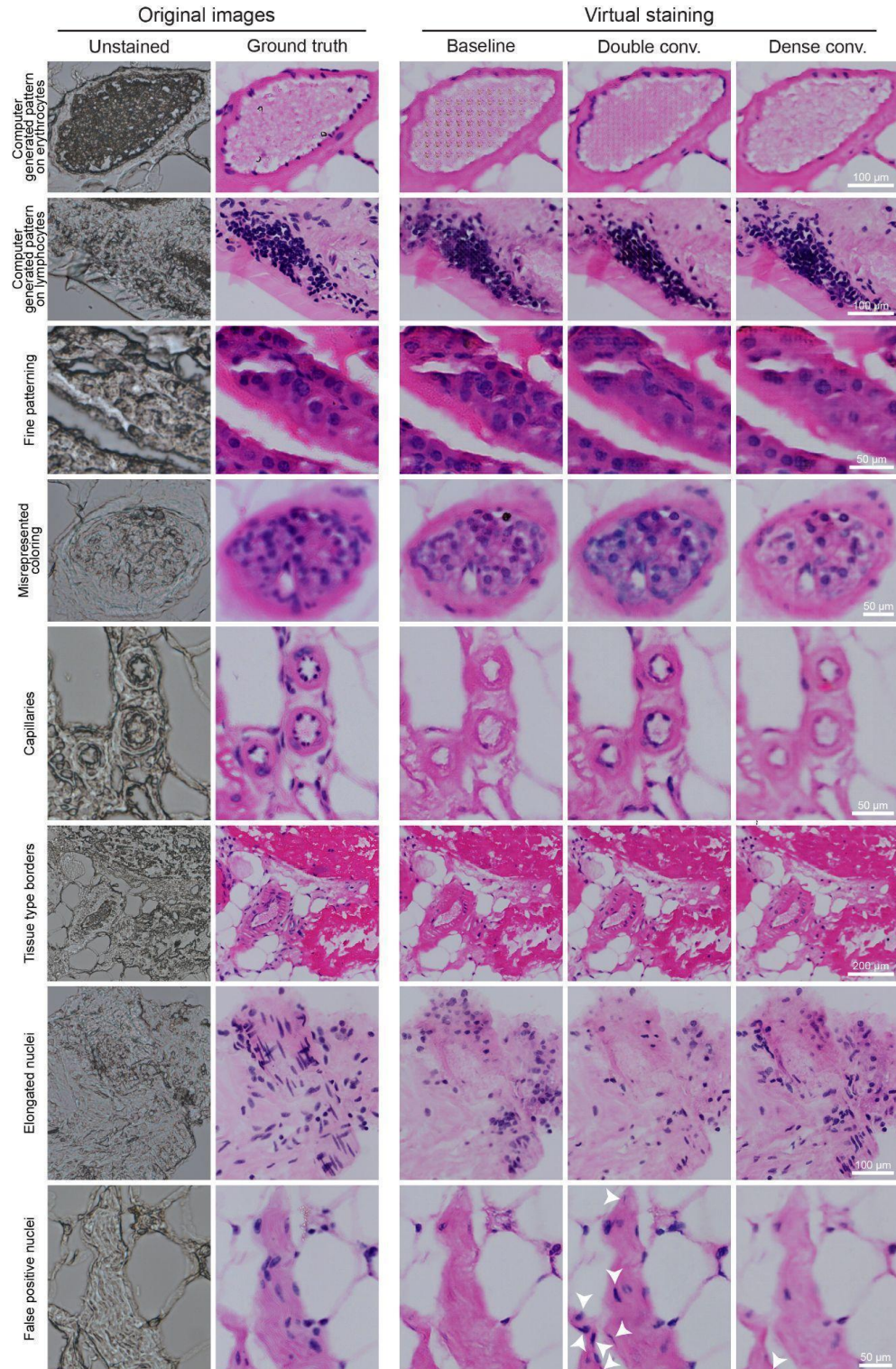


Figure S3. Computation-originated artifacts present in virtually stained tissue images. Each row represents different artifacts, including patterns of different scale, color differences and challenging nuclei. Each column shows original images of unstained or H&E-stained tissue, or virtually stained tissue with baseline, double convolution or dense convolution architecture. Arrowheads indicate false positive nuclei. Scale bars per row.

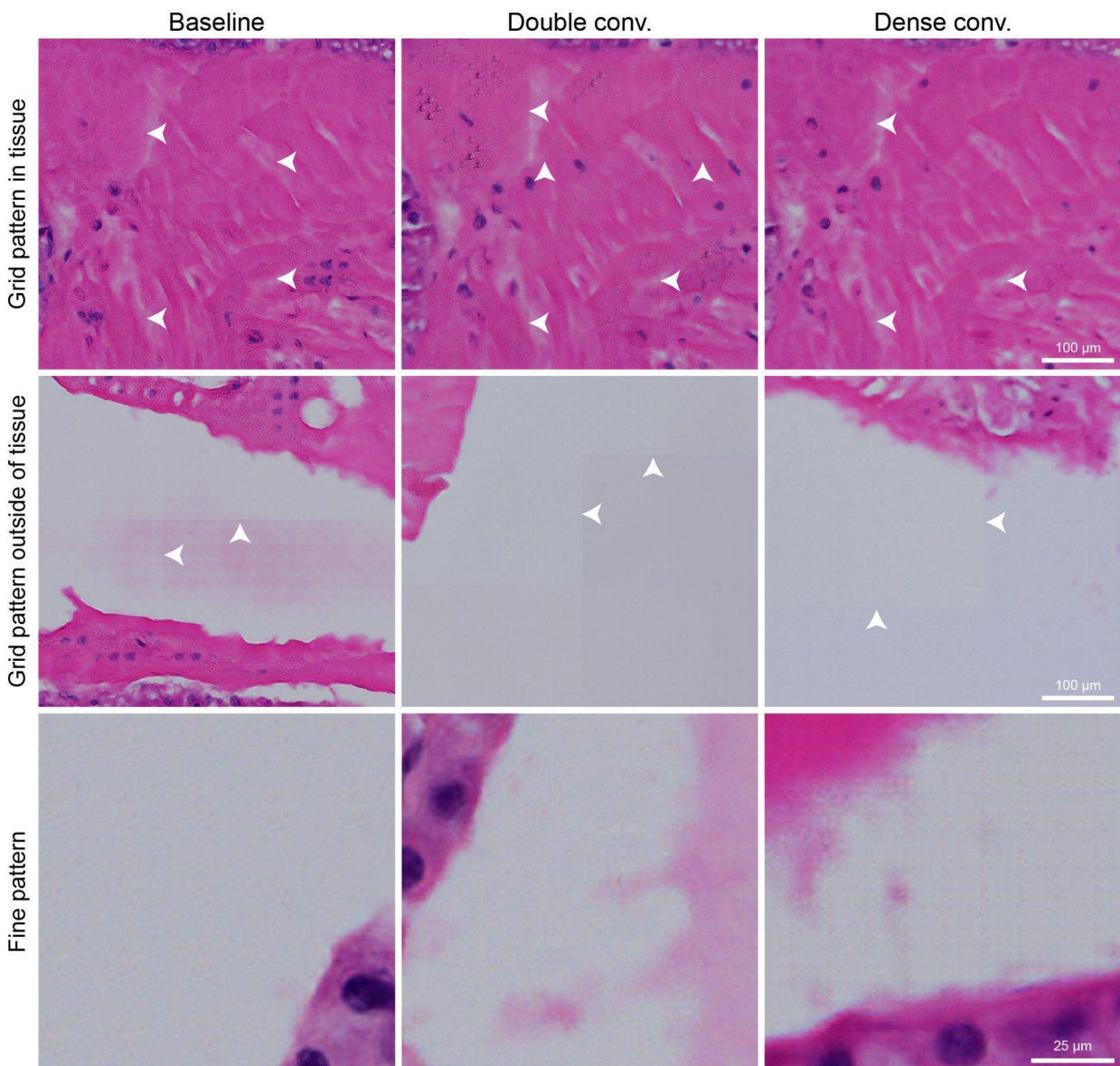


Figure S4. Tiling and pattern artifacts present in virtually stained tissue images. Grid pattern caused by tiling (upper and middle rows) is visible in and outside of tissue. Each of the virtual staining architectures produce a different fine pattern visible in certain tissue images (bottom row). Arrows indicate tile and grid pattern borders. Scale bars per row.

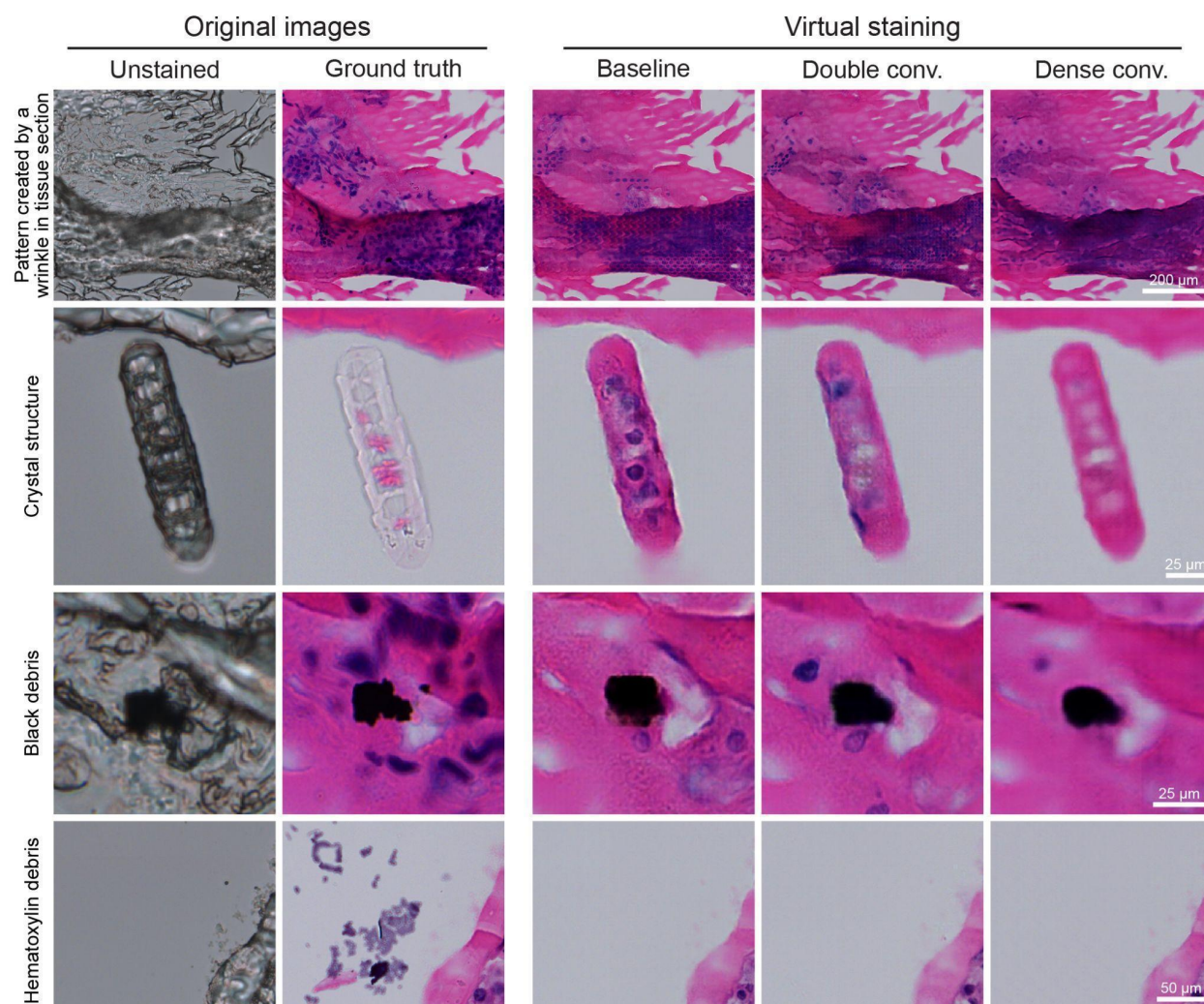


Figure S5. Tissue sample-originated artifacts in virtually stained tissue images. Wrinkles and out-of-focus areas in tissues cause patterning in virtual staining (upper row). Crystal structures displayed false positive staining (second row) whereas black debris was correctly left unstained (third row) by the virtual staining algorithms. Debris from hematoxylin visible in ground truth H&E-stained images was not displayed in virtually stained tissue images (bottom row). Each column shows original images of unstained or H&E-stained tissue, or virtually stained tissue with baseline, double convolution or dense convolution architecture. Scale bars per row.

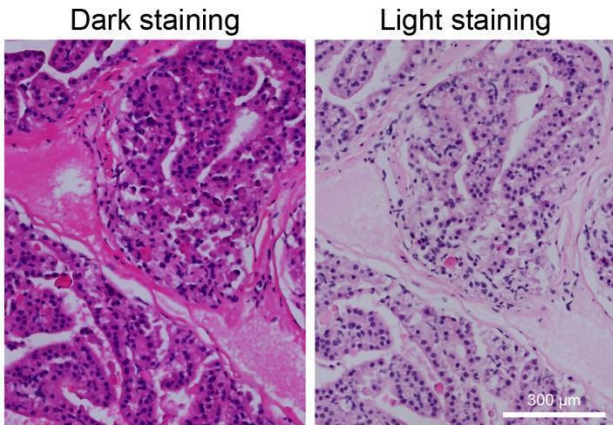


Figure S6. Differences in color manifestation of H&E ground truth. H&E stained ground truth had two batches of tissues which had different staining intensity of eosin.

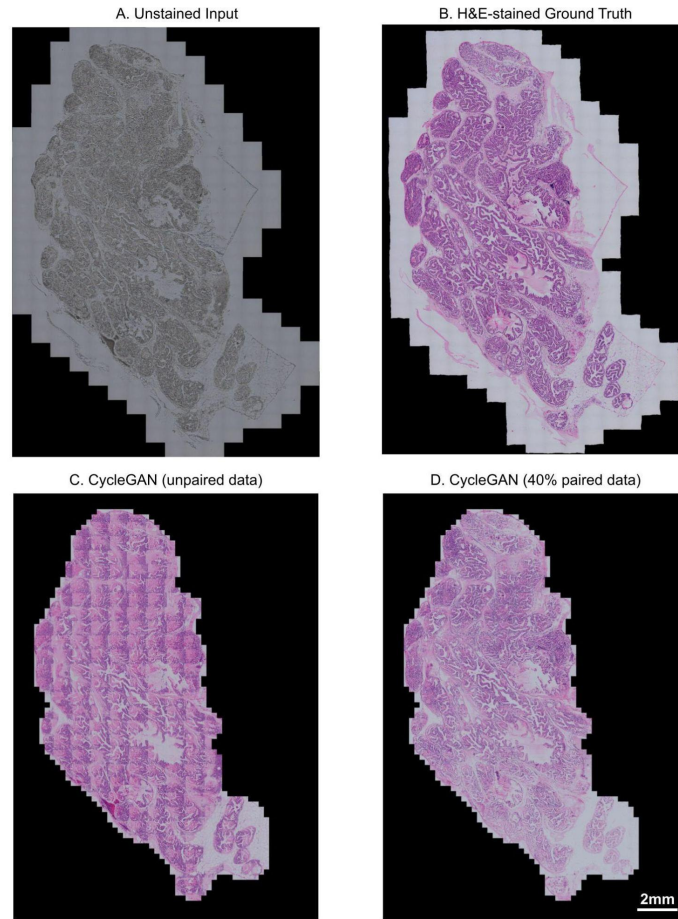


Figure S7. CycleGAN and semi-supervised virtual staining. A ResNet-based implementation of CycleGAN [S1] was used for virtual staining. A) Unstained input. B) H&E-stained ground truth. C) Unsupervised virtual staining by CycleGAN showing coarse tile template-like artifact which is a prevalent problem with this approach. D) Semi-supervised virtual staining with 40% of paired/registered batches showing relatively better results as compared to the completely unsupervised approach. The tile template-like artifacts have completely vanished, but the tissue color starts fading at the bottom of the tissue.

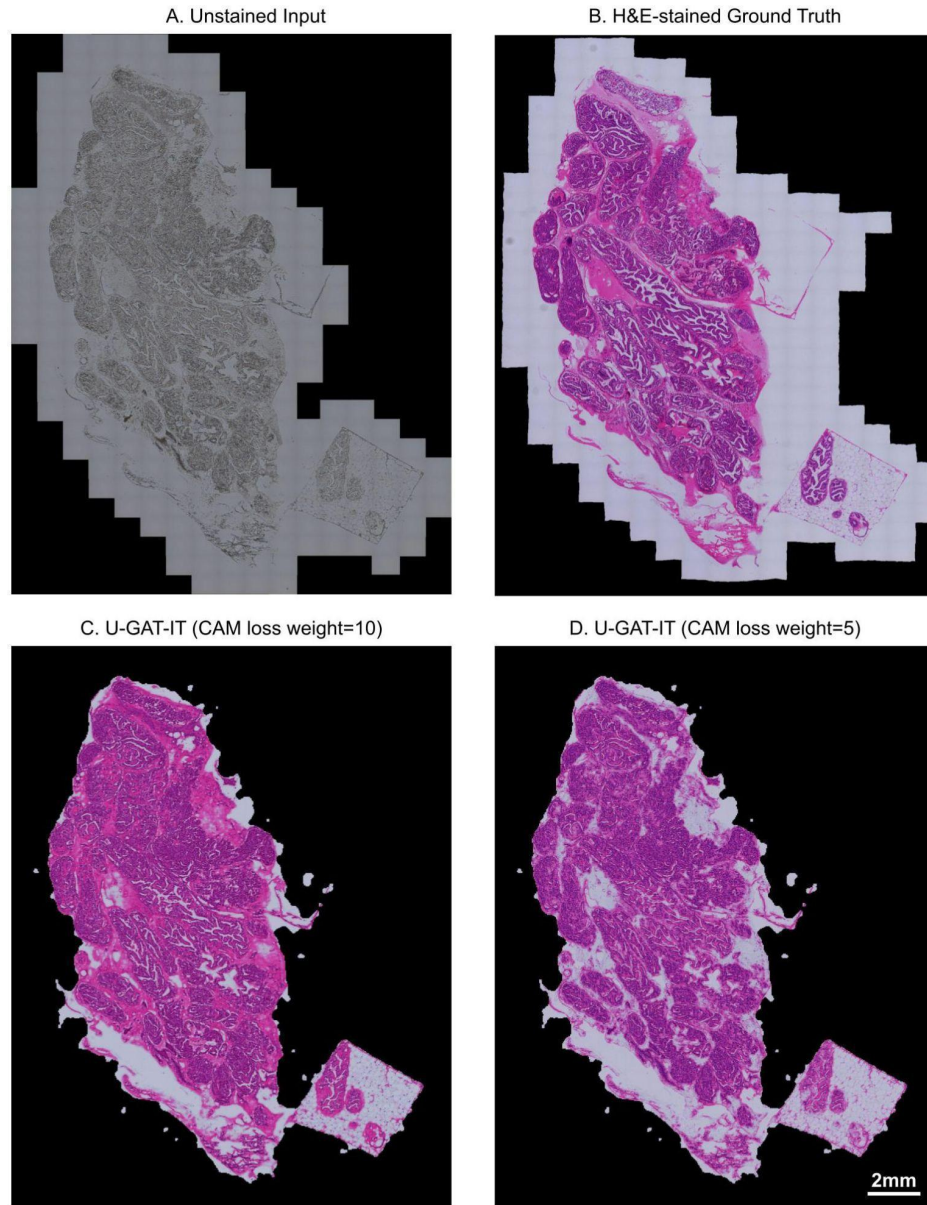


Figure S8. Virtual staining with U-GAT-IT. An unsupervised image-to-image translation method with adaptive layer-instance normalization, attention layers, and class activation map (cam) loss was used for virtual staining [S2]. A) Unstained input. B) H&E-stained ground truth. C) Unsupervised virtual staining using U-GAT-IT trained with loss weights: identity_loss=5 cycle_loss=10 cam_loss=10. Overall staining pattern is reproduced, but fine details are missing from the virtual staining result. D) U-GAT-IT model trained with loss weights: identity_loss=5 cycle_loss=10 cam_loss=5. Virtually stained image has a rough resemblance to the ground truth and the stain looks faded randomly in different regions of the tissue sample.

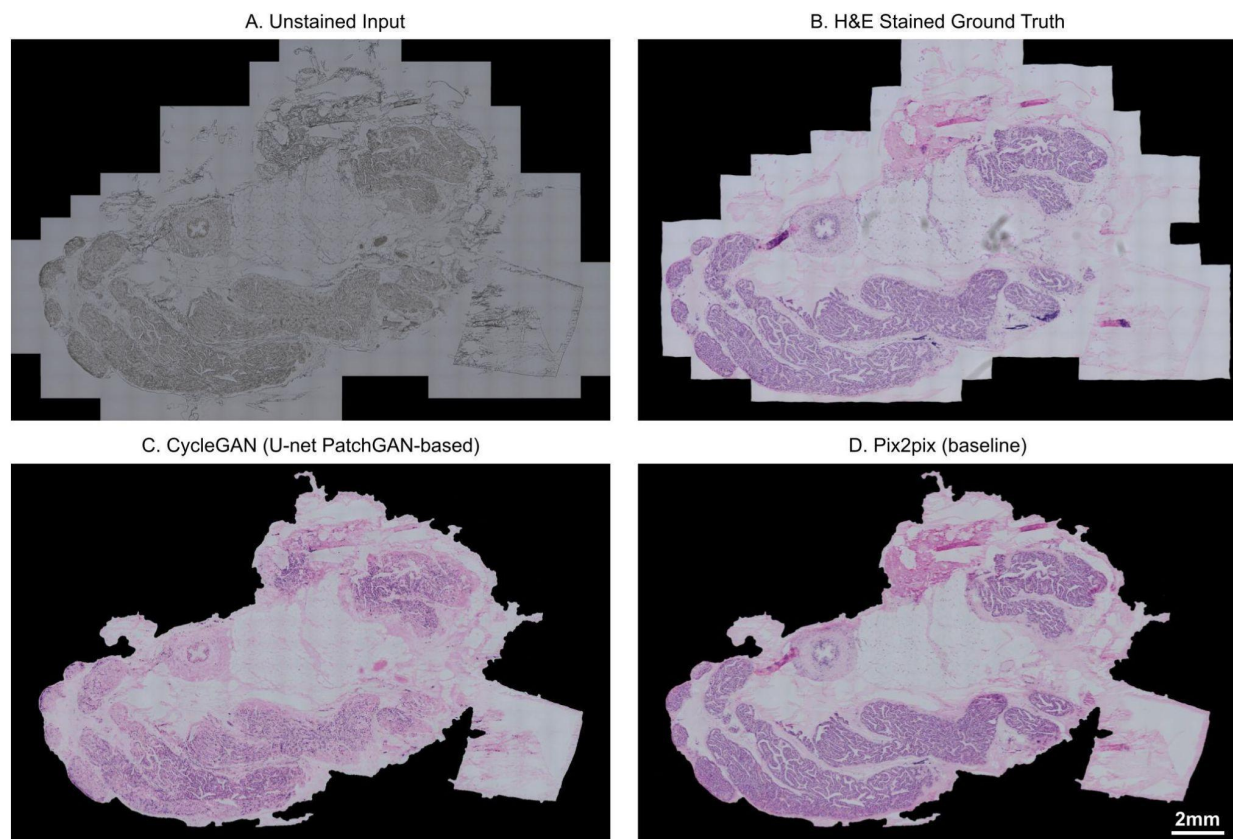


Figure S9. Virtual Staining with CycleGAN and pix2pix. A U-net PatchGAN-based CycleGAN [S1] and pix2pix [S3] (baseline) were trained with four WSIs for the task of virtual staining. A) Unstained input. B) H&E-stained ground truth. C) Virtually stained image using CycleGAN shows relatively better overall quality as compared to other unsupervised methods. D) Virtually stained image using pix2pix has a strong overall resemblance to the ground truth based on visual evaluation.

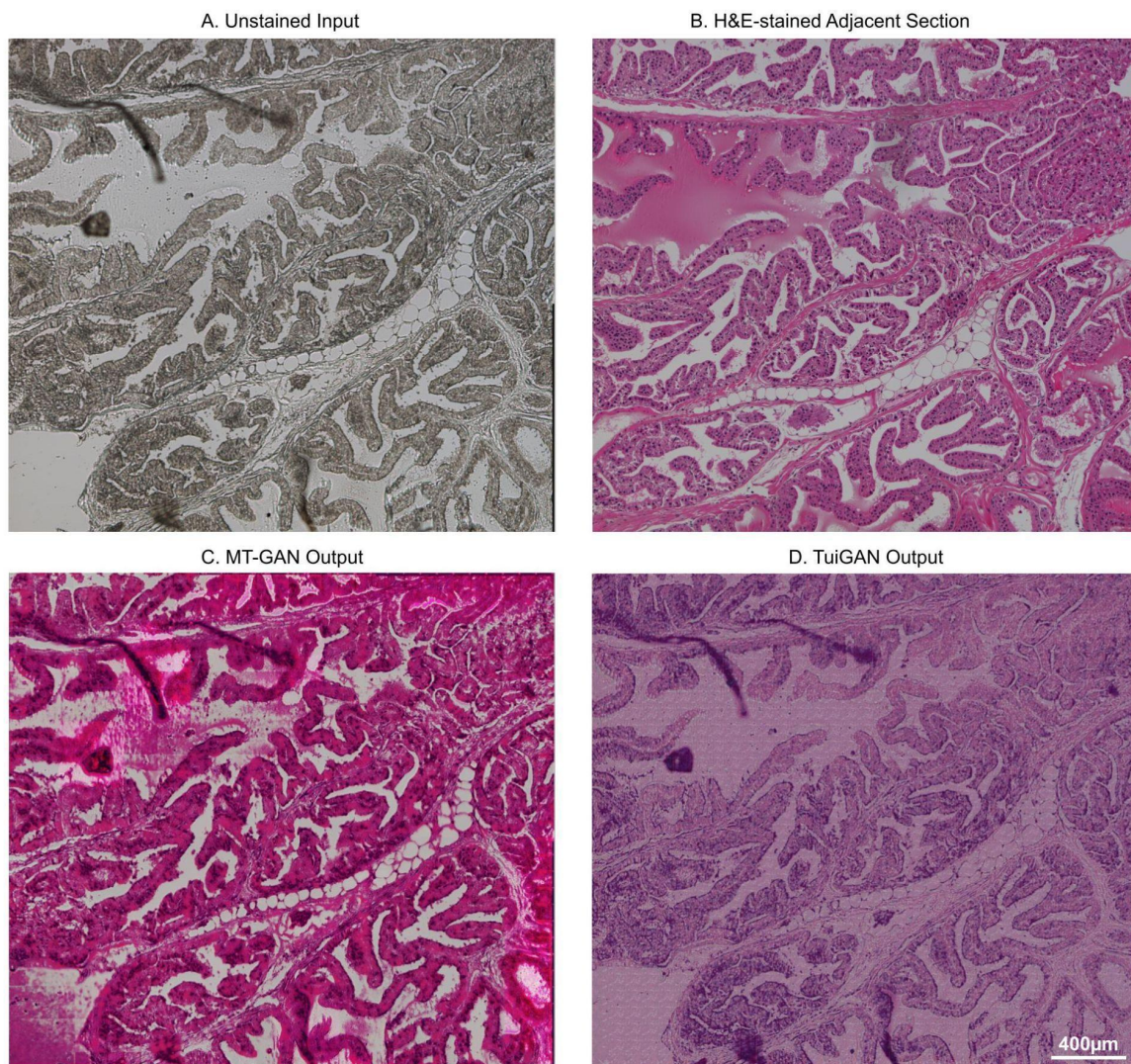


Figure S10. Virtual Staining with one-shot and few-shot methods. TuiGAN [S4], a one-shot learning method, and MT-GAN [S5], a few-shot learning method, were used for virtual staining. A) Unstained input. B) H&E-stained adjacent section. C) MT-GAN virtual staining results from a model previously trained with natural images and fine-tuned with tissue images. Although the macro-level reproduction of the tissue structure looks reasonable, the color is too bright and the placement of nuclei is rather sporadic and random. D) TuiGAN virtual staining results from a model trained with a single tile pair. The resulting image quality is unacceptable for any practical use in this application.

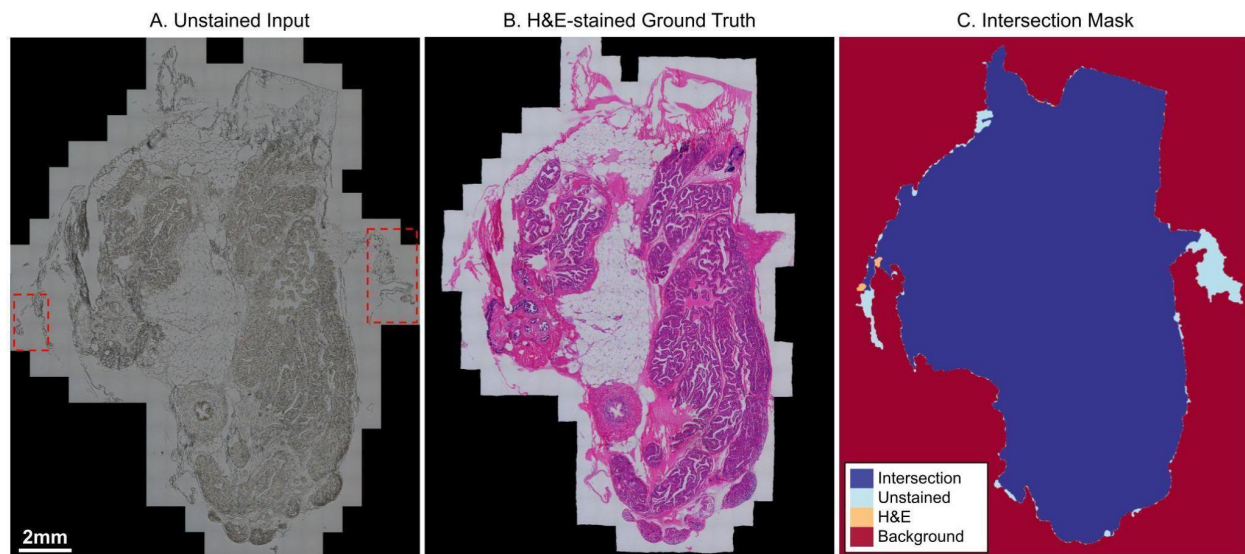


Figure S11. Post-registration intersection mask. Intersection mask to only consider tissue present in both stained and unstained WSI. A) Unstained tissue image. The red dashed ROI represents regions not imaged in the H&E-stained image. B) Registered H&E-stained image. C) A composite representation of unstained, H&E-stained, and intersection masks.

Supplemental Tables

Table S1. Results from virtual staining experiment including edge tiles. Tile-wise comparison of baseline, double convolution, and dense convolution experiments' virtually H&E-stained images against the histochemically H&E stained ground truth images. The evaluation also includes edge tiles. The numbers in bold represent the best results.

WSI	SSIM			PSNR			PCC		
	Baseline	Double Conv.	Dense Conv.	Baseline	Double Conv.	Dense Conv.	Baseline	Double Conv.	Dense Conv.
Sample 1	0.783	0.781	0.784	22.763	22.751	22.958	0.784	0.798	0.801
Sample 2	0.764	0.760	0.767	22.003	21.985	22.216	0.813	0.823	0.825
Sample 3	0.748	0.755	0.761	21.819	22.077	22.419	0.832	0.844	0.848
Sample 4	0.773	0.777	0.783	22.154	22.447	22.916	0.818	0.829	0.836
Sample 5	0.733	0.740	0.747	21.322	21.578	21.750	0.804	0.813	0.818
Sample 6	0.733	0.734	0.755	22.480	21.978	22.134	0.910	0.914	0.919
Sample 7	0.753	0.741	0.771	20.617	19.916	20.330	0.791	0.789	0.795
Sample 8	0.807	0.797	0.826	23.659	23.296	23.827	0.875	0.886	0.881
Sample 9	0.804	0.787	0.825	23.459	23.160	24.031	0.862	0.857	0.866
Sample 10	0.741	0.728	0.774	22.284	21.504	22.561	0.838	0.835	0.849
Sample 11	0.722	0.720	0.753	20.702	20.478	21.297	0.783	0.791	0.794
Sample 12	0.725	0.708	0.753	21.324	20.775	21.838	0.822	0.818	0.831
Sample 13	0.720	0.702	0.749	21.218	20.433	21.496	0.834	0.829	0.845
Mean	0.754	0.748	0.773	21.985	21.721	22.290	0.828	0.833	0.839
Std. Dev.	0.030	0.031	0.032	0.955	1.073	1.258	0.037	0.036	0.0421

Table S2. Lookup Table. Test set sample names to data set file names mapping

Test set sample name	Data set file names
Sample 1	sample_18_unstained.tif
Sample 2	sample_25_unstained.tif
Sample 3	sample_38_unstained.tif
Sample 4	sample_42_unstained.tif
Sample 5	sample_46_unstained.tif
Sample 6	sample_13_unstained.tif
Sample 7	sample_3_unstained.tif
Sample 8	sample_54_unstained.tif
Sample 9	sample_61_unstained.tif
Sample 10	sample_80_unstained.tif
Sample 11	sample_87_unstained.tif
Sample 12	sample_93_unstained.tif
Sample 13	sample_96_unstained.tif

References

1. Zhu, J. Y., Park, T., Isola, P., and Efros, A. A. (2017). Unpaired image-to-image translation using cycle-consistent adversarial networks. In Proceedings of the IEEE international conference on computer vision, pp. 2223-2232. 10.1109/ICCV.2017.244.
2. Kim, J., Kim, M., Kang, H. and Lee, K. (2019). U-gat-it: Unsupervised generative attentional networks with adaptive layer-instance normalization for image-to-image translation. Preprint arXiv:1907.10830.
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4. Lin, J., Pang, Y., Xia, Y., Chen, Z. and Luo, J. (2020). Tuigan: Learning versatile image-to-image translation with two unpaired images. In European Conference on Computer Vision, pp. 18-35. 10.1007/978-3-030-58548-8_2
5. Lin, J., Wang, Y., Chen, Z. and He, T. (2020). Learning to transfer: unsupervised domain translation via meta-learning. In Proceedings of the AAAI Conference on Artificial Intelligence (Vol. 34, No. 07), pp. 11507-11514. 10.1609/aaai.v34i07.6816